

Patient-Reported Outcomes, Treatment Persistence, and Acute Medication Use With Zefranova in Adults With Chronic Migraine: A Global Observational Research Manuscript

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Abstract

Chronic migraine is associated with persistent headache burden, impaired functioning, frequent acute medication use, and reduced quality of life. Beyond migraine-day reduction, patient-reported outcomes, persistence, adherence, and medication-use patterns are important measures of preventive-treatment value. Zefranova, the drug name associated with Zoflevrix, is a simulated once-daily oral preventive therapeutic concept for adults with chronic migraine. To evaluate simulated patient-reported outcomes, persistence, adherence, and acute medication-use changes among adults receiving once-daily oral Zefranova in a global observational research setting. This simulated 36-week prospective observational study included adults with chronic migraine and prior inadequate response to at least one preventive therapy. Participants initiated once-daily oral Zefranova under a structured protocol. Outcomes included monthly migraine days, acute medication-use days, HIT-6 score, MSQ Role Function-Restrictive score, Patient Global Impression of Change, persistence, adherence, and adverse events. Among 486 simulated participants, mean baseline monthly migraine days and acute medication-use days were 16.2 and 12.1. At week 36, monthly migraine days decreased by 5.4 days and acute medication-use days by 3.8 days. A $\geq 50\%$ migraine-day reduction occurred in 39.5% of participants. HIT-6 improved by 7.1 points, MSQ-RFR by 14.8 points, persistence was 78.4%, and mean adherence was 86.7%. Adverse events were mostly mild to moderate. Once-daily oral Zefranova was associated with improved functioning, reduced acute medication use, acceptable persistence, and manageable tolerability. These hypothesis-generating findings require confirmation in prospective real-world studies.

Keywords: Chronic migraine; Zefranova; Zoflevrix; patient-reported outcomes; treatment persistence; adherence; acute medication use; headache impact; migraine prevention; observational study

Introduction

Chronic migraine is a complex neurological disorder characterized by frequent headache days, recurrent migraine-associated symptoms, functional impairment, and substantial reliance on acute medication. Although reduction in monthly migraine days remains a central efficacy outcome in preventive migraine research, it does not fully capture the lived experience of

patients. Many adults with chronic migraine continue to experience impaired work productivity, reduced social participation, sleep disruption, cognitive burden, sensory sensitivity, and emotional distress even when migraine frequency improves.

For neurologists and headache specialists, preventive-treatment evaluation often requires a multidimensional approach. Monthly migraine days, acute medication-use days, headache impact, patient-reported quality of life, adherence, and persistence all contribute to the overall assessment of clinical value. A therapy that reduces migraine frequency but is poorly tolerated or difficult to maintain may have limited long-term usefulness. Conversely, a treatment with moderate efficacy, practical administration, and acceptable tolerability may provide meaningful benefit for patients who need sustained preventive support.

Zefranova, the active drug name associated with Zoflevrix, is represented in this manuscript as a simulated investigational once-daily oral preventive therapeutic concept for adults with chronic migraine. The once-daily oral design is intended to support practical use in patients who may prefer oral therapy or who have experienced inadequate response, tolerability limitations, or discontinuation with prior preventive approaches.

The present simulated observational manuscript evaluates patient-reported outcomes, treatment persistence, adherence, acute medication-use patterns, and safety outcomes among adults with chronic migraine receiving Zefranova over 36 weeks. The objective is to explore a clinically oriented evidence scenario relevant to global neurology and headache-specialist practice.

Methods

Study Design

This was a simulated, prospective, multicenter, observational research manuscript designed for internal scientific-content workflow demonstration. The study followed adults with chronic migraine who initiated once-daily oral Zefranova and were observed for 36 weeks. The design was intended to reflect a pragmatic evidence-generation scenario focused on functional outcomes, treatment persistence, and medication-use patterns.

The study included a baseline assessment period, scheduled follow-up assessments at weeks 4, 12, 24, and 36, and ongoing electronic diary capture of migraine days, headache days, and acute medication use. No real patients, real investigator sites, real ethics approvals, or real trial registration identifiers were used.

Patient Population

The simulated study population included adults aged 18–65 years with chronic migraine and at least four migraine days per month at baseline. Participants had previously experienced inadequate response, tolerability limitations, or discontinuation with at least one preventive therapy. The population was designed to represent adults commonly managed in neurology and headache-specialist settings.

Inclusion Criteria

Participants were eligible for inclusion if they met the following simulated criteria:

- Age 18–65 years.
- Clinical history consistent with chronic migraine.
- Four or more migraine days per month during the baseline assessment period.
- Prior inadequate response, intolerance, or discontinuation of at least one preventive migraine therapy.
- Ability to complete electronic diaries and patient-reported outcome instruments.
- Stable acute medication pattern during the baseline period.
- Willingness to participate in longitudinal follow-up assessments.

Exclusion Criteria

Participants were excluded if they met any of the following simulated criteria:

- Unstable neurological condition other than migraine.
- Clinically significant uncontrolled systemic illness likely to affect study participation.
- Recent initiation of another preventive migraine therapy during the observation window.
- Unstable psychiatric condition that could compromise outcome reporting.
- Prior exposure to Zefranova.
- Pregnancy or lactation.
- Inability to complete electronic diary or follow-up assessments.

Intervention

All participants initiated once-daily oral Zefranova 60 mg. The medication was taken at approximately the same time each day, with or without food. Participants continued permitted acute migraine medications as needed, and acute medication use was recorded through electronic diaries.

Because this was a simulated observational manuscript, there was no randomized comparator group. Findings were summarized as within-participant changes from baseline over time.

Outcomes

The primary outcome was change from baseline to week 36 in Headache Impact Test-6 score.

Secondary outcomes included:

- Change from baseline in monthly migraine days.
- Change from baseline in monthly headache days.
- Change from baseline in monthly acute medication-use days.
- Change from baseline in Migraine-Specific Quality of Life Role Function-Restrictive score.
- Proportion of participants achieving $\geq 50\%$ reduction in monthly migraine days.
- Patient Global Impression of Change at week 36.
- Treatment persistence at weeks 12, 24, and 36.
- Mean adherence over the observation period.

- Incidence and type of treatment-emergent adverse events.

Statistical Analysis

Continuous variables were summarized using means and standard deviations. Categorical variables were summarized using counts and percentages. Within-participant changes from baseline were summarized descriptively at each follow-up time point. No inferential claim of statistical confirmation is made. Missing data were handled using observed-case summaries, with sensitivity analyses based on conservative assumptions for discontinuation and incomplete diary reporting.

Ethical and Data Considerations

This manuscript is based entirely on simulated data. No real patients were enrolled, no real clinical sites participated, and no actual ethics committee approval, informed consent process, or clinical trial registration was involved. All patient-level values, author names, institutional affiliations, and outcomes are fictional and intended only for structured scientific-writing demonstration.

Results

Patient Disposition

A total of 486 simulated participants initiated Zefranova and were included in the observational analysis. At week 12, 443 participants remained on treatment; at week 24, 406 remained on treatment; and at week 36, 381 remained on treatment. The simulated treatment-persistence rate at week 36 was 78.4%.

The most common reasons for discontinuation were insufficient perceived benefit, adverse events, withdrawal from follow-up, and preference for another preventive-treatment strategy. Discontinuation due to adverse events occurred in 6.8% of participants.

Baseline Characteristics

Participants had substantial baseline migraine burden. Mean baseline monthly migraine days were 16.2, and mean monthly headache days were 21.3. The mean baseline acute medication-use frequency was 12.1 days per month. Most participants had previously tried at least one preventive therapy, and nearly half had prior exposure to two or more preventive classes.

Table 1. Baseline Demographic and Clinical Characteristics

Characteristic	Overall Population (N = 486)
Mean age, years	42.6
Female, n (%)	352 (72.4%)
Mean duration of migraine history, years	14.1
Mean monthly migraine days	16.2
Mean monthly headache days	21.3

Characteristic	Overall Population (N = 486)
Mean monthly acute medication-use days	12.1
Prior inadequate response to ≥ 1 preventive therapy, n (%)	486 (100.0%)
Prior exposure to ≥ 2 preventive classes, n (%)	231 (47.5%)
Baseline HIT-6 score, mean	65.1
Baseline MSQ-RFR score, mean	46.9
Comorbid sleep disturbance, n (%)	174 (35.8%)
Comorbid anxiety symptoms reported, n (%)	139 (28.6%)

HIT-6 = Headache Impact Test-6; MSQ-RFR = Migraine-Specific Quality of Life Role Function-Restrictive domain.

Primary Outcome: Headache Impact

At week 36, mean HIT-6 score improved by 7.1 points from baseline. Improvements were observed progressively over time, with a mean reduction of 3.2 points at week 12, 5.8 points at week 24, and 7.1 points at week 36. These simulated results suggest a meaningful reduction in headache-related impact among participants who initiated Zefranova and remained engaged in follow-up.

Migraine Frequency and Acute Medication Use

Monthly migraine days decreased from a baseline mean of 16.2 to 10.8 at week 36, corresponding to a mean reduction of 5.4 days. Monthly headache days decreased by 6.2 days, and monthly acute medication-use days decreased by 3.8 days.

Table 2. Changes in Migraine Burden and Patient-Reported Outcomes

Outcome	Baseline Mean	Week 12 Change	Week 24 Change	Week 36 Change
Monthly migraine days	16.2	-3.1	-4.7	-5.4
Monthly headache days	21.3	-3.8	-5.6	-6.2
Monthly acute medication-use days	12.1	-2.0	-3.3	-3.8
HIT-6 score	65.1	-3.2	-5.8	-7.1
MSQ-RFR	46.9	+7.4	+12.3	+14.8

Outcome	Baseline Mean	Week 12 Change	Week 24 Change	Week 36 Change
score				

The proportion of participants achieving a $\geq 50\%$ reduction in monthly migraine days at week 36 was 39.5%. A $\geq 30\%$ reduction was observed in 58.2% of participants. Improvements in migraine frequency were generally accompanied by reduced acute medication use, although the magnitude of reduction varied across participants.

Patient-Reported Global Improvement

At week 36, 61.7% of participants reported that their migraine condition was “much improved” or “moderately improved” on the Patient Global Impression of Change scale. A smaller proportion reported no meaningful change, and a limited number reported worsening. Participants who reported global improvement generally also showed larger reductions in monthly migraine days and acute medication use.

Table 3. Patient Global Impression of Change at Week 36

Patient-Reported Category	Participants, n (%)
Much improved	128 (26.3%)
Moderately improved	172 (35.4%)
Minimally improved	83 (17.1%)
No change	57 (11.7%)
Minimally worse	24 (4.9%)
Moderately worse	15 (3.1%)
Much worse	7 (1.4%)

Treatment Persistence and Adherence

Treatment persistence was 91.2% at week 12, 83.5% at week 24, and 78.4% at week 36. Mean adherence over the observation period was 86.7%, based on simulated tablet accountability and diary confirmation. Among participants who persisted through week 36, 84.1% maintained adherence of at least 80%.

Table 4. Treatment Persistence and Adherence

Measure	Week 12	Week 24	Week 36
Participants remaining on treatment, n	443	406	381
Treatment persistence, %	91.2%	83.5%	78.4%
Mean adherence, %	89.4%	87.8%	86.7%
Adherence $\geq 80\%$, n (%)	409 (84.2%)	394 (81.1%)	366 (75.3%)

Persistence appeared to be influenced by perceived early benefit, tolerability, patient expectation, and routine integration of once-daily dosing. Participants reporting meaningful improvement by week 12 were more likely to remain on treatment through week 36.

Safety and Tolerability

Treatment-emergent adverse events were reported by 266 participants. Most were mild or moderate. The most common adverse events were nausea, fatigue, dizziness, somnolence, and dry mouth. Adverse events leading to discontinuation were reported in 33 participants.

Table 5. Safety and Tolerability Summary

Safety Outcome	Overall Population (N = 486)
Any treatment-emergent adverse event, n (%)	266 (54.7%)
Mild adverse event, n (%)	171 (35.2%)
Moderate adverse event, n (%)	82 (16.9%)
Severe adverse event, n (%)	13 (2.7%)
Serious adverse event, n (%)	6 (1.2%)
Discontinuation due to adverse event, n (%)	33 (6.8%)
Nausea, n (%)	57 (11.7%)
Fatigue, n (%)	51 (10.5%)
Dizziness, n (%)	39 (8.0%)
Somnolence, n (%)	37 (7.6%)
Dry mouth, n (%)	34 (7.0%)

No simulated pattern of progressive laboratory abnormality, severe hypersensitivity, or major cardiovascular imbalance was observed. Because the dataset is simulated and the study duration was limited to 36 weeks, uncommon or delayed risks cannot be evaluated.

Discussion

This simulated observational manuscript examined patient-reported outcomes, treatment persistence, adherence, acute medication use, and safety among adults with chronic migraine receiving once-daily oral Zefranova. The simulated findings suggest improvement across multiple clinically relevant domains, including headache impact, migraine frequency, acute medication use, and migraine-specific role functioning.

The reduction in HIT-6 score was selected as the primary outcome because headache impact is a patient-centered measure that reflects daily burden beyond attack frequency alone. A mean improvement of 7.1 points by week 36 suggests that participants experienced meaningful functional improvement in this simulated dataset. Improvements in MSQ-RFR score further suggest reduced restriction in daily roles and activities.

Migraine-frequency reduction remained clinically relevant, with a mean decrease of 5.4 monthly migraine days by week 36. This degree of reduction may be meaningful for patients with high baseline burden, particularly when accompanied by reduced acute medication use. In chronic migraine management, acute medication-use reduction is important because frequent use may complicate disease management and contribute to ongoing headache burden in some patients.

Treatment persistence is an important real-world consideration. In this simulated population, week-36 persistence was 78.4%, which suggests that once-daily oral administration and acceptable tolerability may support continued use for many patients. However, persistence should not be interpreted as a direct measure of efficacy. Patients may discontinue for many reasons, including inadequate perceived benefit, adverse events, access issues, cost, treatment expectations, comorbidities, or preference for another preventive strategy.

The simulated safety profile was generally manageable. Nausea, fatigue, dizziness, somnolence, and dry mouth were the most frequently reported adverse events. These events are relevant to clinical decision-making because they may influence adherence, daily functioning, and willingness to continue preventive therapy. Discontinuation due to adverse events occurred in 6.8% of participants, indicating that tolerability monitoring would remain important.

For neurologists and headache specialists, the simulated findings emphasize the value of evaluating preventive therapy through both clinical and patient-centered measures. A patient may experience modest reduction in migraine days but meaningful improvement in functioning, acute medication reliance, or treatment confidence. Conversely, a frequency reduction may not translate into acceptable treatment value if adverse effects or adherence barriers are substantial.

Clinical Implications for Neurologists and Headache Specialists

The simulated results support several practical considerations for specialist care. First, patient-reported outcomes should be included when evaluating preventive-treatment response because migraine burden is multidimensional. Second, reduction in acute medication-use days may be an important complementary endpoint, particularly for patients with frequent rescue-medication use. Third, adherence and persistence should be assessed proactively rather than assumed, even with once-daily oral therapy.

In a clinical setting, Zefranova could be positioned as a preventive-treatment concept for adults with chronic migraine who require an oral option after inadequate response or tolerability limitations with prior preventive therapy. Treatment discussions should include expected time to response, realistic goals, adverse-event monitoring, acute medication-use tracking, and follow-up planning.

Specialists may also consider structured response assessment at 12 weeks and 24 weeks, incorporating migraine-day frequency, acute medication use, HIT-6 or similar impact measures, patient-reported functional improvement, and tolerability. Such an approach may support individualized decisions regarding continuation, dose adjustment, or transition to alternative preventive strategies.

Limitations

This manuscript has several important limitations. All data are simulated and should not be interpreted as real clinical evidence. No real patients were enrolled, and no actual treatment exposure occurred. Therefore, the results cannot establish efficacy, effectiveness, safety, comparative value, regulatory status, or real-world clinical performance.

The observational design lacks a randomized comparator group, making it unsuitable for causal inference. Improvements could reflect regression to the mean, expectation effects, changes in

acute medication behavior, improved diary engagement, natural fluctuation in migraine frequency, or other unmeasured factors. The simulated dataset also does not capture complex real-world barriers such as medication access, affordability, comorbid psychiatric disease, polypharmacy, or variable follow-up adherence.

The study duration of 36 weeks may not be sufficient to evaluate long-term persistence, rare adverse events, or durability beyond the observation period. Subgroup analyses by medication-overuse status, aura status, sex, age group, prior therapy class, and psychiatric comorbidity were not fully explored. Future real-world studies would require predefined analytic methods, validated outcome capture, representative populations, and appropriate comparator frameworks.

Conclusion

In this simulated global observational manuscript, once-daily oral Zefranova was associated with improvements in headache impact, monthly migraine days, acute medication-use days, migraine-specific role functioning, treatment persistence, and adherence among adults with chronic migraine. The simulated tolerability profile was generally manageable, with mostly mild to moderate adverse events.

These findings are hypothesis-generating and intended only for internal scientific-content workflow demonstration. They do not represent real-world clinical evidence for Zoflevrix or Zefranova. Prospective clinical studies and rigorously designed real-world evidence programs would be required to evaluate the true efficacy, safety, persistence, and patient-reported value of Zefranova in chronic migraine prevention.

Data Transparency Statement

All data presented in this manuscript are simulated. No real patient data, clinical trial records, investigator-site information, ethics approvals, or regulatory documents were used. The dataset was generated only to demonstrate structured medical-writing content for a fictional research scenario.

Funding Statement

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Conflicts of Interest

All authors, institutions, affiliations, and disclosures are fictional. No real financial relationships, consulting arrangements, employment relationships, stock ownership, honoraria, advisory roles, or conflicts of interest are represented.

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